

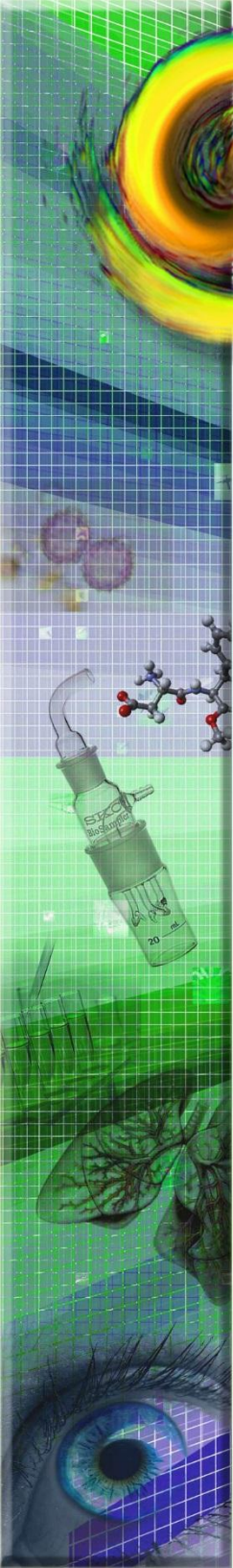
ARE Labs Inc.

Title: Foaming Cleanser Spray Coverage Testing

Client: Bragg & Bevier LLC

Project #: 11048.10

1/29/2026



Testing Overview

- Purpose of testing was to determine the spray coverage of the product inside a 96-gallon trash bin.
- 4 height levels in the trash bin were selected, including the interior lid surface
- Mounted 16 1"x1" paper filter coupons (4 per height including 4 on interior lid surface) to 96-gallon bin interior walls using adhesive.
- Initial mass of the spray can, and test coupons were recorded.
- Placed aerosol can upright at the center of the bin floor using double sided tape to secure it.
- Installed internal lighting and video recording device.
- Actuated can for full discharge with the lid closed.

$$\text{Surface Deposition (mg/in}^2\text{)} = \frac{\Delta\text{Mass}}{\text{Area}}$$

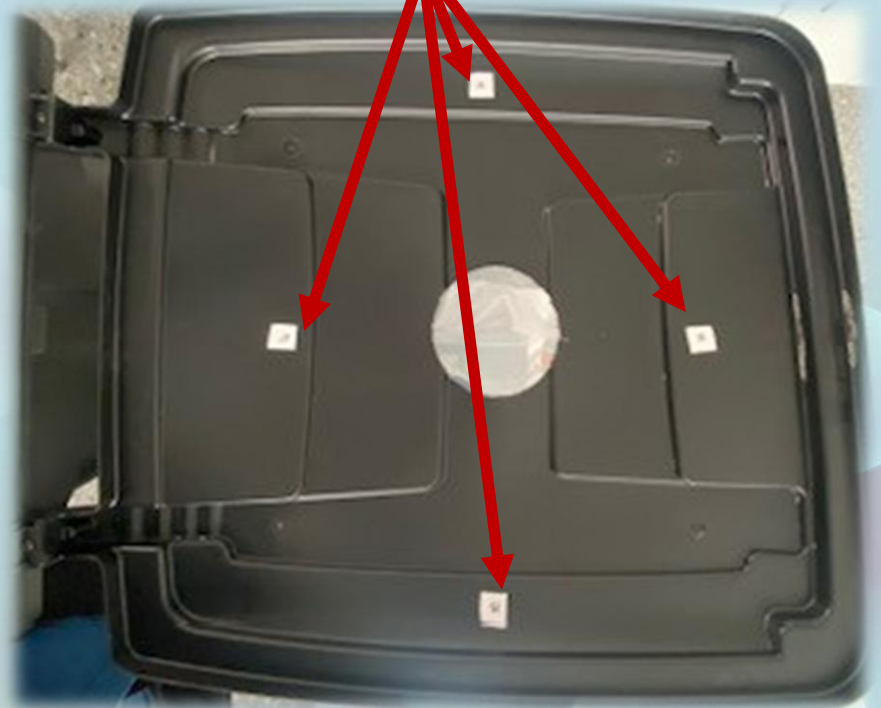
Testing Overview

- Allowed foam to settle for 3 minutes.
- Post-testing mass of the coupons was taken, and the surface deposition was calculated.
- Removed coupons and placed them in labeled containers.
- Post-testing mass of the spray can was recorded.



Configuration

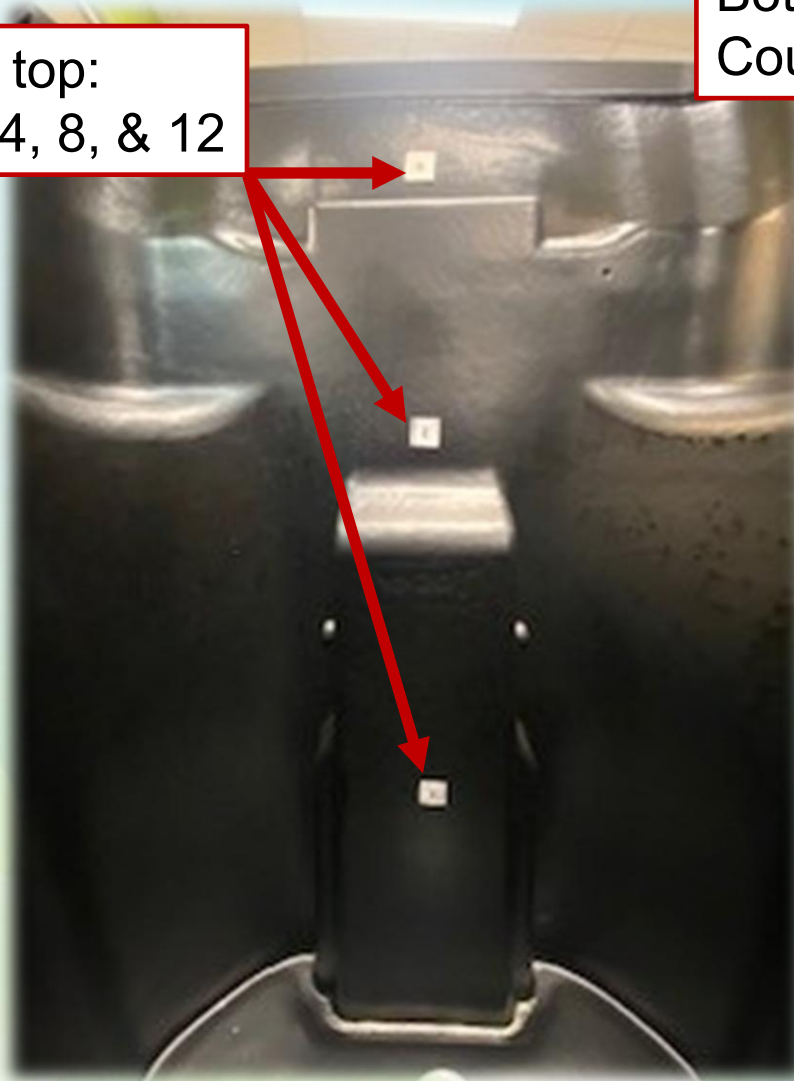
Coupons 13-16



Configuration

Front

Bottom to top:
Coupons 4, 8, & 12



Back

Bottom to top:
Coupons 2, 6, & 10



Configuration

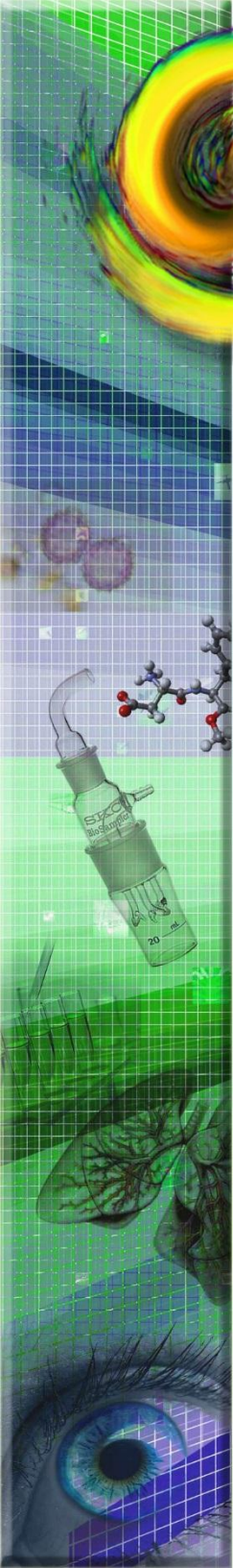
Left

Bottom to top:
Coupons 1, 5, & 9

Bottom to top:
Coupons 3, 7, & 11

Right

Bottom to top: 1ft,
2ft, & 3ft
markings



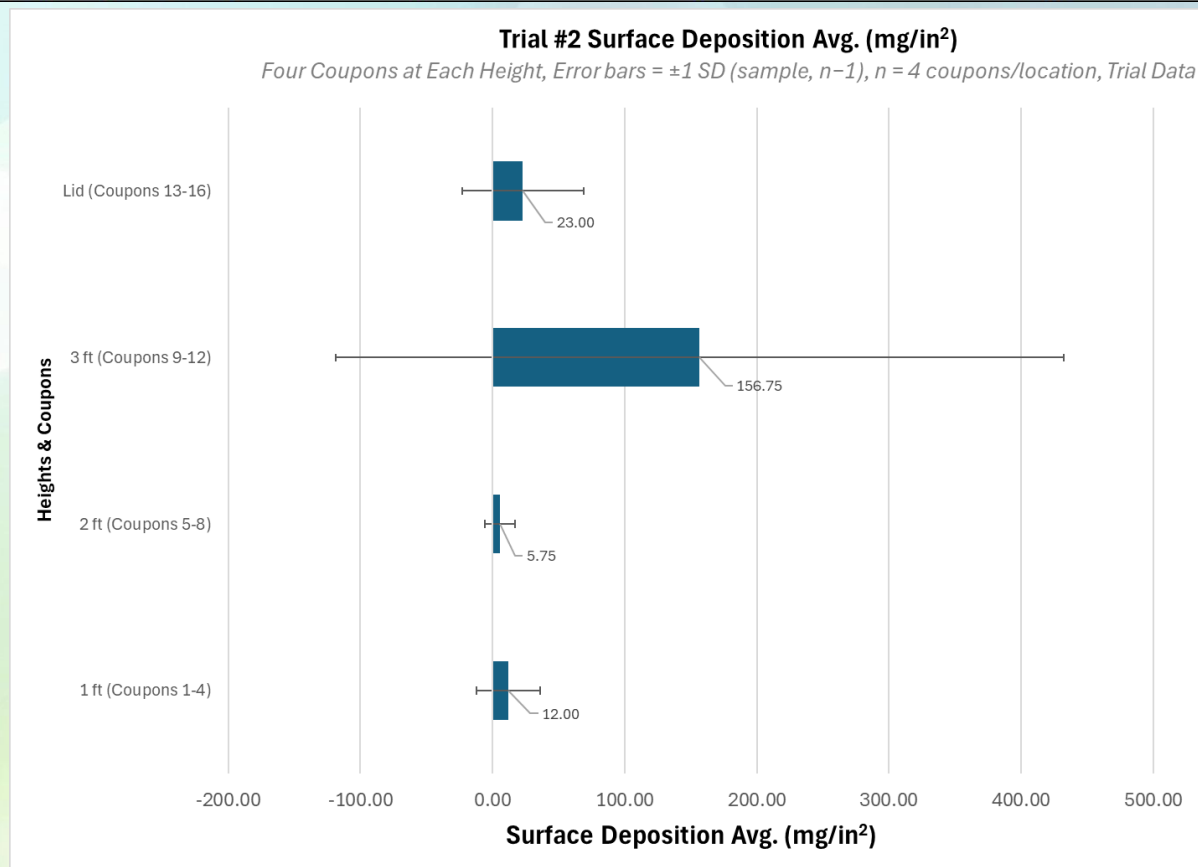
Foaming Cleanser Spray Coverage Results

Trial #2 Results

4 coupons at each height.

Spray Can Mass			
Trial #	Initial Mass of Spray Can (no lid)	Post-Testing Mass	Spray Formula Mass
Trial #1	131.51g	77.29g	54.22g
Trial #2	156.82g	65.18g	91.64g

Coupon Mass Average Trial #2			
# ID	Initial Mass Avg. (mg)	Post-testing Mass Avg. (mg)	Surface Deposition Avg. (mg/in ²)
1 ft (Coupons 1-4)	64.25	76.25	12.00
2 ft (Coupons 5-8)	65.00	70.75	5.75
3 ft (Coupons 9-12)	63.00	219.75	156.75
Lid (Coupons 13-16)	71.50	94.50	23.00

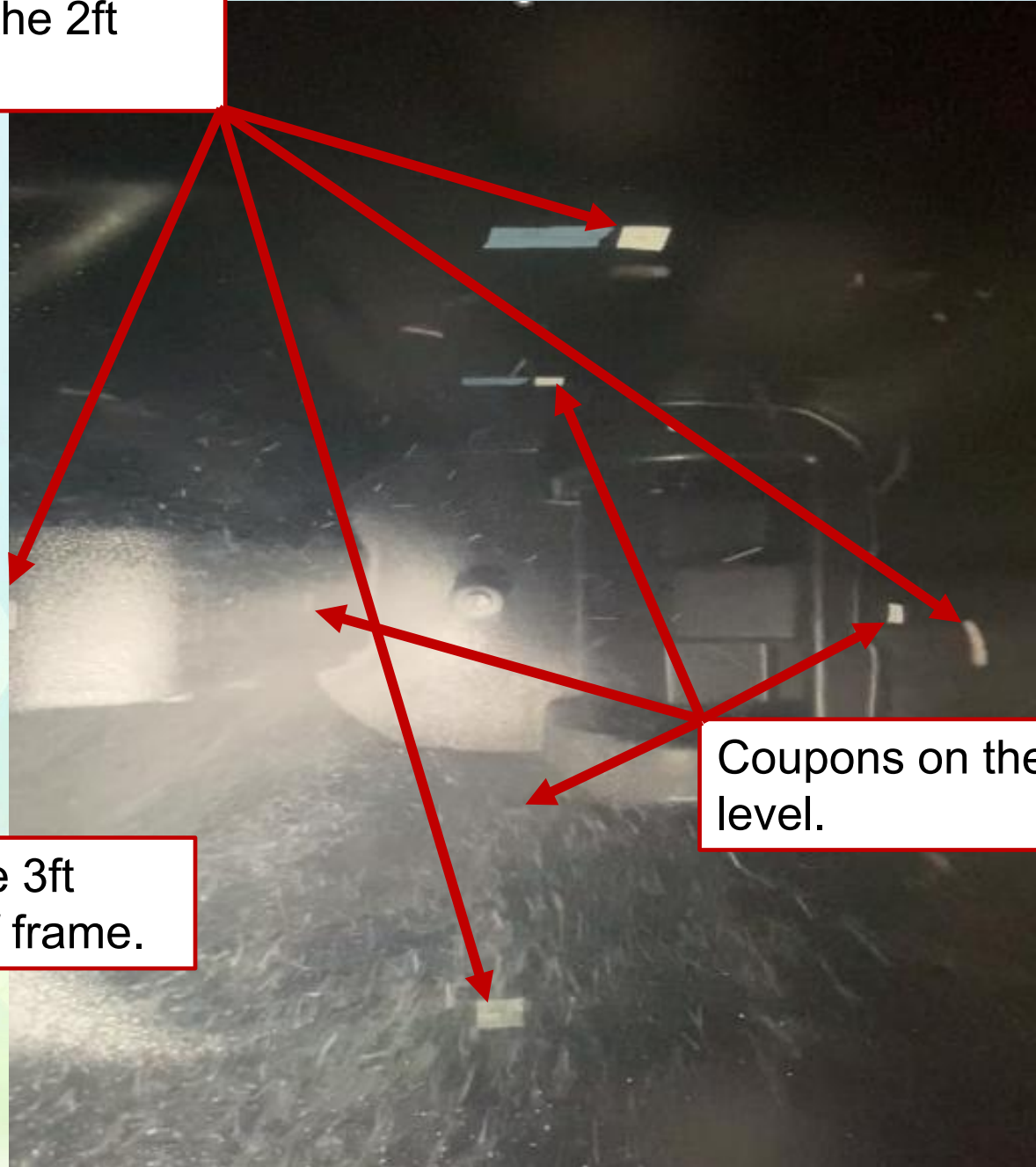


Trial # 2 Mid-Point

Coupons on the 2ft level.

Coupons on the 1ft level.

Coupons at the 3ft level are out of frame.



Trial #2 Mid-Point



Since the spray was fan shaped it mainly only covered one side and hardly anything was got outside what is displayed in the circle.

Trial #3 Results

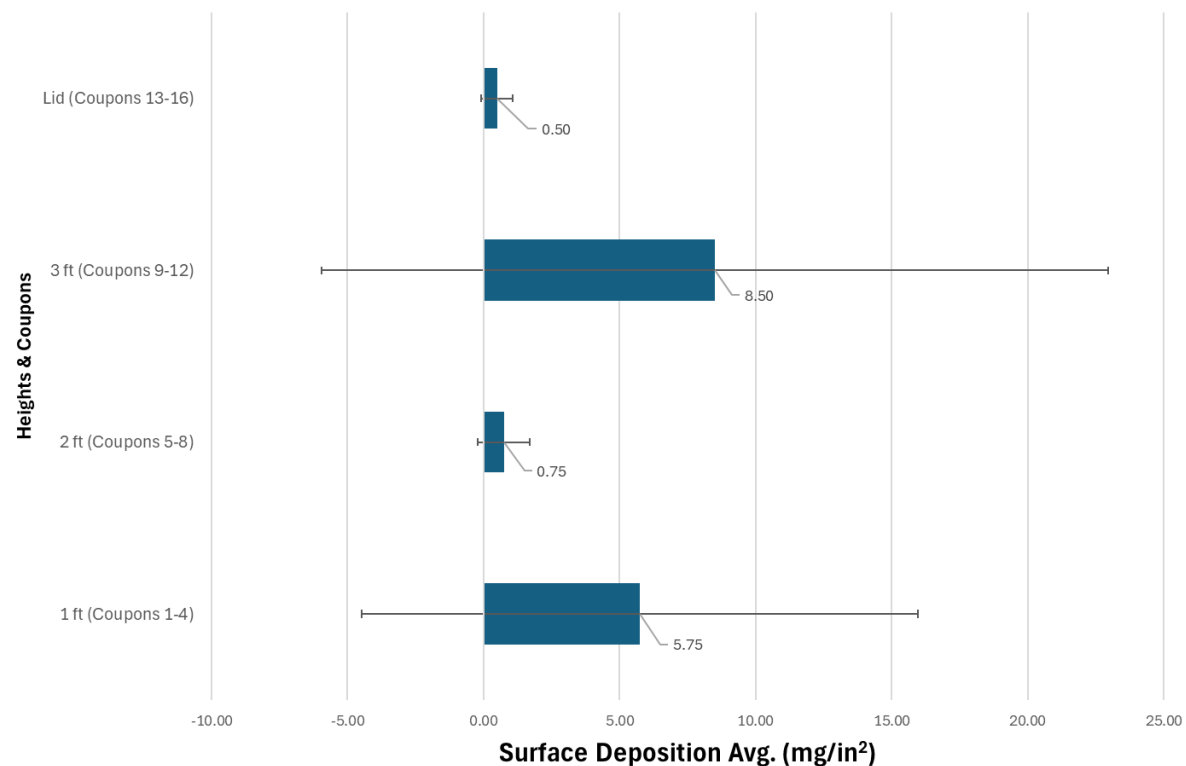
4 coupons at each height.

Spray Can Mass			
Trial #	Initial Mass of Spray Can (no lid) (g)	Post-Testing Mass (g)	Spray Formula Mass (g)
Trial #3	149.94	63.98	85.96
Trial #4	162.27	64.58	97.69
Trial #5	135.40	91.48	43.92

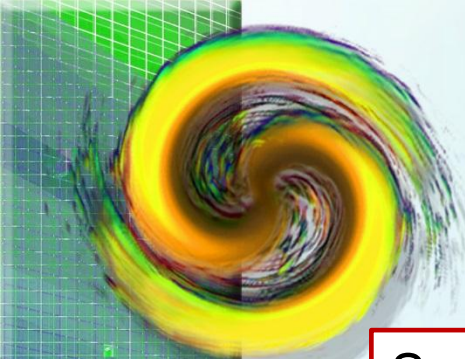
Coupon Mass Average Trial #3			
# ID	Initial Mass Avg. (mg)	Post-testing Mass Avg. (mg)	Surface Deposition Avg. (mg/in ²)
1 ft (Coupons 1-4)	62.00	67.75	5.75
2 ft (Coupons 5-8)	68.50	69.25	0.75
3 ft (Coupons 9-12)	73.50	82.00	8.50
Lid (Coupons 13-16)	76.50	77.00	0.50

Trial #3 Surface Deposition Avg. (mg/in²)

Four Coupons at Each Height, Error bars = ± 1 SD (sample, $n-1$), $n = 4$ coupons/location, Trial Data



Trial #3 Mid-Point



Spray can

The spray for most of the time was spraying with a constant stream but went to a full cone shape when 10% of the contents inside remained.



Trial #3 End-Point

The latch on the nozzle when pushed down angles the spray toward it and as a result the spray can mostly went towards the corner that is indicated.



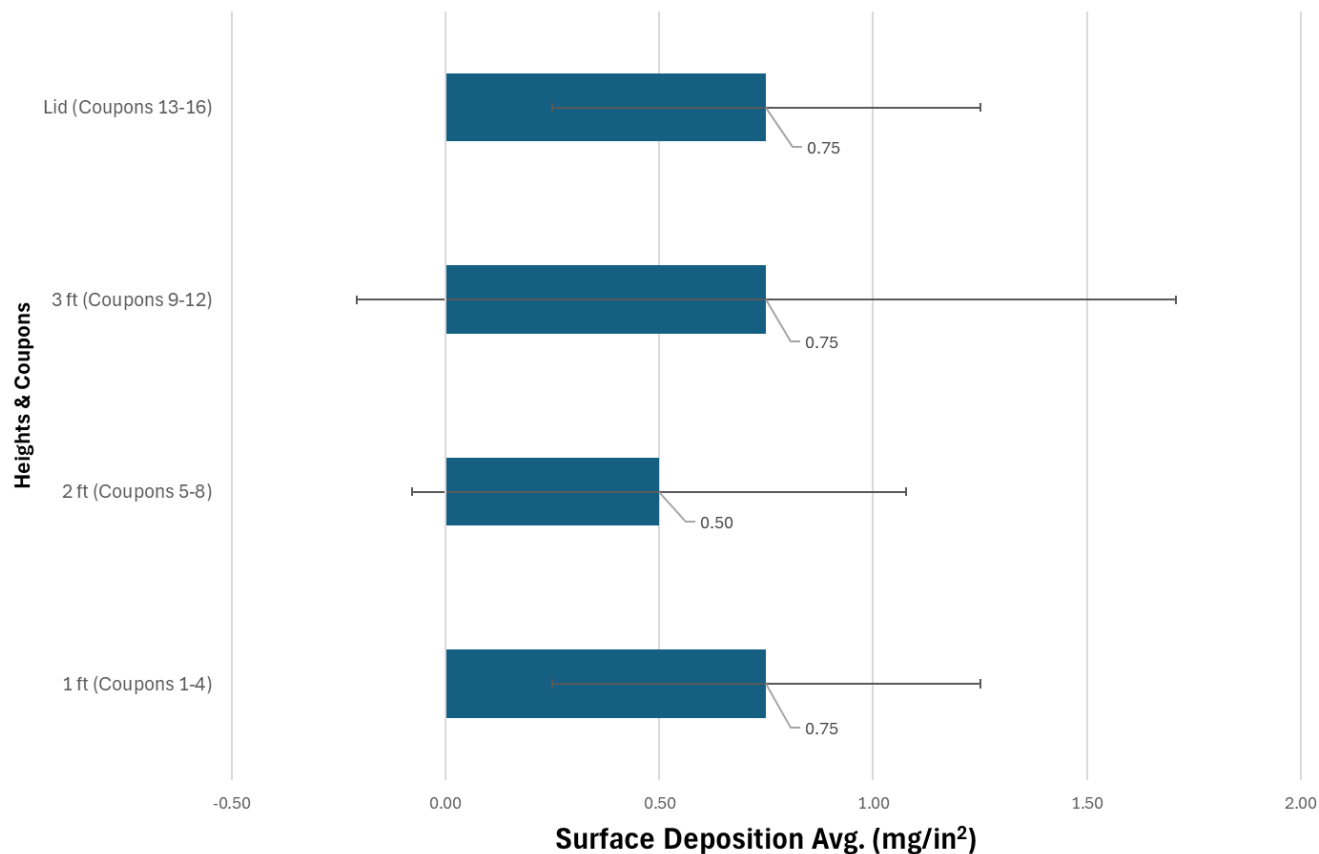
Trial #4 Results

4 coupons at each height.

Coupon Mass Average Trial #4			
# ID	Initial Mass Avg. (mg)	Post-testing Mass Avg. (mg)	Surface Deposition Avg. (mg/in ²)
1 ft (Coupons 1-4)	72.25	73.00	0.75
2 ft (Coupons 5-8)	66.00	66.50	0.50
3 ft (Coupons 9-12)	68.25	69.00	0.75
Lid (Coupons 13-16)	65.75	66.50	0.75

Trial #4 Surface Deposition Avg. (mg/in²)

Four Coupons at Each Height, Error bars = ± 1 SD (sample, $n-1$), $n = 4$ coupons/location, Trial Data



Trial #4 Mid-Point

The spray for most of the time was spraying with a constant stream but went to a full cone shape when 10% of the contents inside remained.



Trial #4 End-Point

The latch on the nozzle when pushed down angles the spray toward it and as a result the spray can mostly went towards the corner that is indicated.



Trial #5 Results

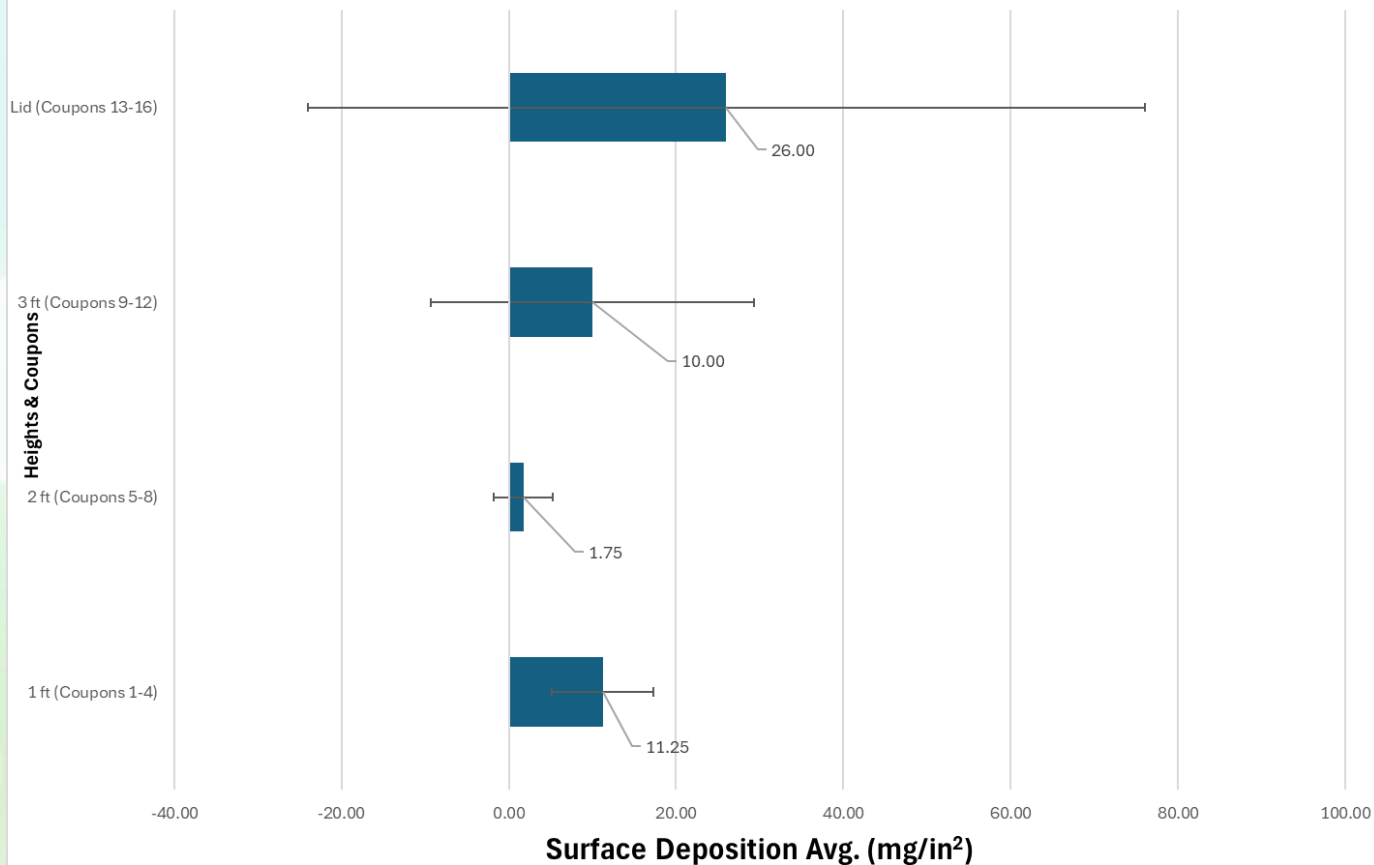
4 coupons at each height.

Coupon Mass Average Trial #5

# ID	Initial Mass Avg. (mg)	Post-testing Mass Avg. (mg)	Surface Deposition Avg. (mg/in ²)
1 ft (Coupons 1-4)	66.50	77.75	11.25
2 ft (Coupons 5-8)	65.25	67	1.75
3 ft (Coupons 9-12)	70.25	80.25	10.00
Lid (Coupons 13-16)	65.75	91.75	26.00

Trial #5 Surface Deposition Avg. (mg/in²)

Four Coupons at Each Height, Error bars = ± 1 SD (sample, $n-1$), $n = 4$ coupons/location, Trial Data



Trial #5 Mid-Point

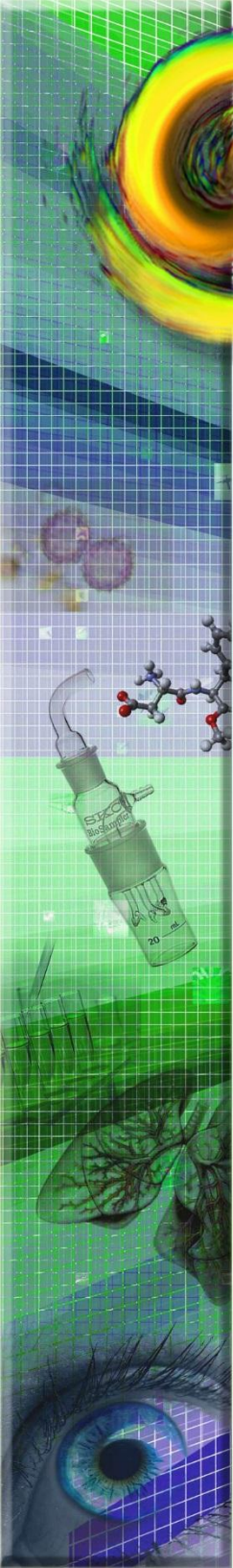
The spray for most of the time was spraying with a constant stream but went to a full cone shape when 10% of the contents inside remained.



Trial #5 End-Point

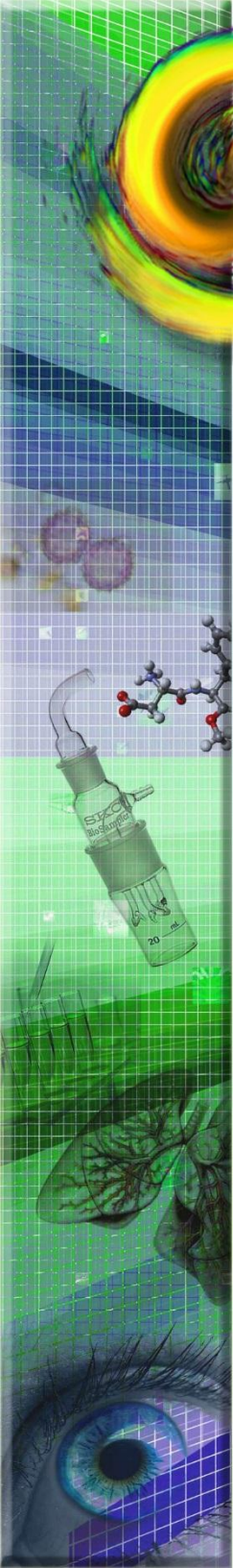
The latch on the nozzle when pushed down angles the spray toward it and as a result the spray can mostly went towards the corner that is indicated.





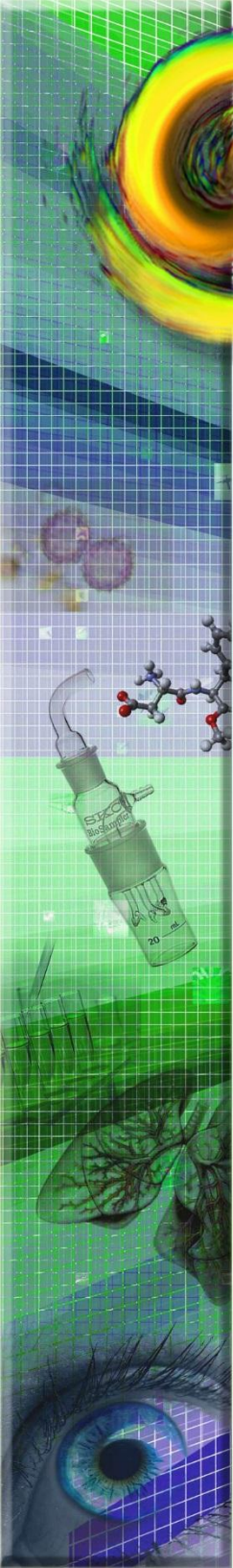
Trial 1&2 Summary

- The 2nd spray can had more output than the can used in the 1st attempt.
- The can sprayed high enough to reach the interior lid surface.
- The contents inside the can had a nice peppermint smell.
- The fan shaped spray covered half the trash can.
- Since the can sprays in a fan shape 5 of the 16 coupons were the only ones to get wet. 1 at the 1ft level, 1 at the 2ft level, 2 at the 3ft level, and 1 at the lid.



Trial 3-5 Summary

- The three new samples with the updated nozzle was tested.
- The new samples that had the nozzle modified sprayed higher than the two initial spray cans that were tested.
- The can sprayed high enough to reach the interior lid surface.
- The contents inside the can had a nice peppermint smell.
- All tested cans discharged primarily as a single stream through most of can life and transitioned to the intended full-cone only near depletion (~final 10% remaining).
- Since the cans spray in an angle towards the latch when actuated, they covered half the trash can.



Discussion

- We noticed the spray mostly shoots in a constant stream until it gets close to empty (around 10% left inside) and then it starts spraying in a full cone shape as intended.
- Spray plume was biased in a consistent direction due to the latch on the nozzle being angled when the spray can is activated, leading to preferential coverage of one side of the bin.
- This is likely due to high actuation pressure which directly creates the two failure modes observed.
- This can be fixed with valve selection (lower spring rate), smoother latch mechanics, and reduced friction so the user doesn't need to push as hard.